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ARTHROPODS AND MEDICAL ENTOMOLOGY - ORIGINAL PAPER

Circulation of *Rickettsia* species and rickettsial endosymbionts among small mammals and their ectoparasites in Eastern Slovakia

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Abstract

Bacteria belonging to the genus *Rickettsia* are known as causative agents of vector-borne zoonotic diseases, such as spotted fevers, epidemic typhus and endemic typhus. Different species of ticks, mites and fleas could act as reservoirs and arthropod vectors of different pathogenic *Rickettsia* species. The aim of this work was to establish active surveillance of *Rickettsia* spp. in mites, ticks and fleas collected from small mammals (rodents and shrews) in Eastern Slovakia. A total of 964 animal ear biopsies, 871 mites, 667 ticks and 743 fleas were collected from small mammals in the Košice region, Eastern Slovakia. All specimens were identified using specialized taxonomic keys, and were conserved in ethanol until DNA extraction was performed. After DNA extraction, identification of *Rickettsia* species was performed by PCR-based methods. The total prevalence of rickettsiae from ear biopsies was 4.6% (95% CI, 3.2–5.9), in tested mites 9.3% (95% CI, 7.4–11.2), 17.2% (95% CI, 14.3–20.1) in *I. ricinus* ticks and 3.5% (95% CI, 2.2–4.8) in fleas. Sequence analysis of the partial *gltA* gene and *Rickettsia helvetica*-, *Rickettsia slovaca*-, *Rickettsia raoultii*- species specific real-time PCR tests revealed the presence of *R. helvetica*, *R. slovaca*, unidentified *Rickettsia* and rickettsial endosymbionts. These pathogenic and symbiotic species were confirmed in the following ectoparasite species—*Laelaps jettmari*, *Haemogamasus nidi*, *Laelaps agilis* and *Eulaelaps stabularis* mites, *Ixodes ricinus* ticks, *Ctenophthalmus solutus*, *C. assimilis* and *Megabothris turbidus* fleas infesting host—*Apodemus agrarius*, *A. flavicollis*, *Microtus arvalis* and *Myodes glareolus* small mammals. These results confirm the circulation of *R. helvetica*, *R. slovaca*, unidentified *Rickettsia* and rickettsial endosymbionts in mites, ticks and fleas collected on small mammals in the Košice region, Eastern Slovakia.

Keywords *Rickettsia* · Small mammals · Ticks · Fleas · Mites · Eastern Slovakia

Introduction

A wide range of mammals, including rodents, act as reservoir hosts and can participates in the natural cycle of several zoonotic agents. They play an important role in the circulation of hantaviruses (Klempa et al. 2005), agents of larval toxocarosis (Antolová et al. 2004; Reiterová et al. 2013), leptospirosis (Stanko et al. 1996), chlamydiosis (Čisláková et al. 2004), borreliosis (Hanincová et al. 2003), anaplasmosis (Blaňarová et al. 2014), bartonellosis (Kraljik et al. 2016), babesiosis (Blaňarová et al. 2016) and other zoonosis (Čondlová et al. 2018; Danišová et al. 2017; Mácová et al. 2018). Small mammals are hosts to different ectoparasites such as ticks (Nosek et al. 1972; Labuda et al. 1989; Krasnov et al. 2007, 2010), mites (Kocianová 1980; Stanko 1989), fleas (Rosický 1957; Stanko et al. 2002; Krasnov et al. 2006, 2010) and lice (Stanko et al. 2015). Several small mammals such as *Apodemus flavicollis* (Melchior, 1834), *Apodemus agrarius* (Pallas,

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1771), *Myodes glareolus* (Schreber, 1780), *Microtus arvalis* (Pallas, 1778), *Microtus subterraneus* (de Sélys-Longchamps, 1836), *Micromys minutus* (Pallas, 1771) and *Sorex araneus* (Linnaeus, 1758) have a relevant epidemiological role as hosts to a broad spectrum of ectoparasites mentioned above in Slovakia (Meerburg et al. 2009; Špitalská et al. 2015, 2017; Minichová et al. 2017).

Ticks are obligate ectoparasites that feed on a wide range of mammalian, avian and reptilian hosts. In Europe, changes in tick occurrence are observed as a result of climatic changes, economic and socio-economic conditions. Such changes results in the establishment of new foci and the spread of vector-borne diseases (Bullová et al. 2009; Lukáň et al. 2010). Five genera and 21 species of soft and hard ticks were recognized in different fauna of Slovakia (Slovák 2010) from which epidemiologically important species belonging to *Ixodes ricinus* Linnaeus 1758; *Haemaphysalis concinna* Koch 1844; *Haemaphysalis punctata* Canestrini & Fanzago, 1878; *Haemaphysalis inermis* Birula, 1895; *Dermacentor reticulatus* Fabricius, 179; and *Dermacentor marginatus* Sulzer, 1776. *Ixodes ricinus* collected from vegetation and animals in different locations of Slovakia were previously found to be infected with *Rickettsia helvetica* and *Rickettsia monacensis* (Minichová et al. 2017; Radzijevskaja et al. 2015; Špitalská et al. 2014, 2016, 2018; Švehlová et al. 2014), and *Dermacentor* species with *Rickettsia slovaca* and *Rickettsia raoultii* (Švehlová et al. 2014; Špitalská et al. 2012, 2018). The genus *Rickettsia* includes obligate intracellular Gram-negative bacteria that are responsible for mild to severe diseases in humans, with worldwide distribution (Raoult and Roux 1997). The *Rickettsia* species (Proteobacteria: Rickettsiales) found in Slovakia are pathogenic to humans (Uchiyama 2012; Oteo and Portillo 2012) with mild clinical manifestations such as fever, rash and eschar with different severities (Faccini-Martínez et al. 2014).

About 75 species of fleas (*Siphonaptera*) have been recorded in Slovakia, while more than 30 species were reported from small mammals (Cyprich et al. 2002; Dudich 1993; Rosický 1957; Stanko et al. 2002). Fleas play a role in maintaining and transferring several pathogenic agents such as *Bartonella henselae* causing cat scratch disease and other *Bartonella* species (Brinkerhoff et al. 2010; Marie et al. 2006; Morick et al. 2009, 2010). The oriental rat flea (*Xenopsylla cheopis*) is primarily the vector of *Rickettsia typhi* and transmits murine typhus to humans. Interestingly, some other species of fleas, including the cat flea *(Ctenocephalides felis)*, have been found to be naturally infected with *R. typhi* (Azad and Beard 1998). In Slovakia, the role of fleas in the circulation of rickettsiae is still not well understood. To this day, *Rickettsia felis* and *Rickettsia africae* were detected in Slovakia (Sekeyová et al. 2012; Špitalská et al. 2015). *Rickettsia felis* was identified in *Ctenophthalmus solutus* collected from *A. agrarius.* The cat flea, *Ctenocephalides felis*, is known as the primary vector of *R. felis*, which is the etiological agent of the flea-borne spotted fever with clinical symptoms similar to malaria and dengue in humans. However, it appears to be milder than other rickettsioses (Raoult and Parola 2007). *R. africae* was identified in *Ceratophyllus garei* fleas collected from reed warblers (*Acrocephalus scirpaceus*) migrating from Africa (Sekeyová et al. 2012), However, *R. africae* is usually transmitted by ticks.

Mites have a broad range of eating habits. In fact, they occupy every feeding niche-from predators, phytophagous, decomposers and fungivores (e.g. oribatid mites, or some mesostigmatic mites) to scavengers or parasites of other animals (Badiertakis et al. 2014). Mites of the family *Laelapidae* (Acari: Mesostigmata) are distributed worldwide and are the most common ectoparasites of small rodents (Guo et al. 2013; Karg 1971, 1993; Kocianová 1989; Mašán and Fenďa 2010; Miťková et al. 2015; Rynkiewicz et al. 2013). Mesostigmatic mites are probably involved in the circulation of some pathogens such as the human pathogenic tick-borne encephalitis virus or various rickettsial agents (Kocianová et al. 1993; Miťková et al. 2015; Radzijevskaja et al. 2018; Reeves et al. 2006). The mites of the genus *Allodermanyssus* may act as a vector and reservoir of *Rickettsia akari*, which produced urban outbreaks, with the house mouse *Mus musculus* as the main host for this mite species (Raoult and Roux 1997).

Ectoparasitic mites, fleas and ticks share the body of the mammalian host, and by co-feeding, some uninfected parasites can become infected. In this study, we aimed to investigate the presence and prevalence of *Rickettsia* species and rickettsial endosymbionts in ectoparasites parasitizing small mammals in Slovakia. We predicted (i) the presence of *Rickettsia* species in all studied ectoparasites with the highest prevalence in ticks, (ii) a higher prevalence in the most abundant small mammals species carrying such ectoparasites and (iii) the identification of several rickettsial species.

Materials and methods

Origin of the samples

Study sites were located in sylvatic mixed forests (Hýľov; 500–750 m above sea level (asl); 48° 44′N, 21° 4′E and Čermeľ; 208–420 m asl; 48° 45′N, 21° 8′E), in suburban deciduous forests (Botanical garden, Košice; 208 m asl; 48°44′N, 21° 14′E), as well as in an agricultural site comprising of patches of oak-hornbeam forests and a few cultivated fields and meadows (Rozhanovce 215 m asl; 48° 45′N, 21° 21′E) in Eastern Slovakia (Fig. 1).

In total, 7194 ectoparasites (2635 ticks, 1664 fleas and 2895 mites) were removed from 991 trapped small mammals. Hosts were trapped and ectoparasites were collected between 2011 and 2013. Rodents were trapped alive using Swedish bridge metal traps following the protocol of Stanko (1994),

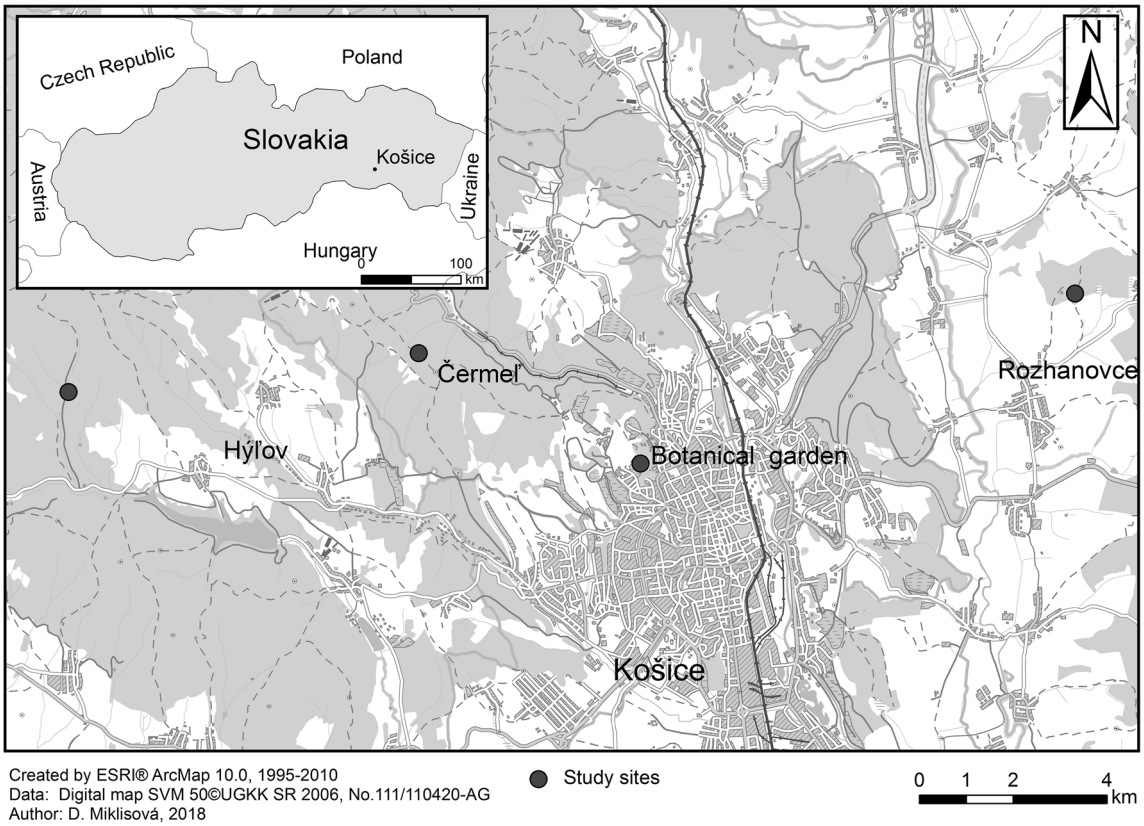


Fig. 1 Map of collection sites in Eastern Slovakia

and Stanko and Miklisova (1995). Rodent trapping was carried out using 100–150 traps per site for a two-night trapping period. Captured rodents were transported to the laboratory, anaesthetized and sacrificed humanely according to current laws of the Slovak Republic. The rodents were identified at species, sex and age class (subadult, adult) levels. Ectoparasites were collected from infested animals using forceps and stored in 70% ethanol until identification. The species and sex of ectoparasites were determined by microscopic examination according to Rosický (1957), Filippova (1977) and Mašán and Fenďa (2010) and stored in 70% ethanol until molecular analysis. Ear skin punch biopsies were stored at − 80 °C.

DNA extraction and molecular analysis

Ectoparasites were washed with sterile water, dried, transferred to individual tubes and crushed with a sterile carbon steel surgical scalpel blade (Surgeon, JAI Surgicals Ltd., India). Total DNA of fleas and mites was isolated by boiling individual specimen in 0.7 M ammonium hydroxide (Rijpkema et al. 1996). Genomic DNA from fed ticks and rodent tissues were extracted using DNeasy Blood and Tissue Kit (Qiagen) following the manufacturer's recommendations. The quantity and quality of DNA were assessed with a NanoPhotometer Pearl (Implen, Germany). DNA samples were stored at − 20 °C and later used as templates for the PCR amplifications. Ectoparasite and tissue samples were screened by PCR-based methods for the presence of *Rickettsia* spp. Identification of *Rickettsia* spp. was performed using a nested PCR based on the genus-specific variable regions of the *gltA* gene using the primer sets RpCS.877p–RpCS.1258n and RpCS.896p–RpCS.1233n (Choi et al. 2005; Regnery et al. 1991). The DNAs from *Rickettsia*-free ticks and nuclease-free water were used as negative controls for each reaction set. DNAs of *R. slovaca* and *R. helvetica* originated from the sample bank of the Institute of Virology BMC SAS Bratislava were used as positive controls. PCR amplifications were conducted using SuperHot Master Mix (Bioron, Germany) as recommended by the manufacturer using a thermocycler Labcycler (SensoQuest, Germany). PCR products were analyzed by electrophoresis in a 1.5% agarose gel, stained with GelRed™ (Biotium, Hayward, California) and visualized with a UV transilluminator. Some *Rickettsia*-positive samples were screened for the presence of *R. helvetica*, *R. slovaca* and *R. raoultii* species using a TaqMan PCR assay targeting a fragment of the 23S rRNA and *ompB* genes, respectively, as previously described (Boretti et al. 2009; Jiang et al. 2012). Randomly selected amplicons of *gltA* gene were purified and analyzed by sequencing both DNA strands. The sequencing was performed by Macrogen (http://www.macrogen.com). Obtained DNA sequences were compared with available

sequences in the GenBank database using the basic local alignment search tool (BLAST) on http://blast.ncbi.nlm.nih.gov/. A phylogenetic analysis was further performed using the MEGA5 software (Tamura et al. 2011).

Statistical analysis

Chi-square and Fisher's tests were used to analyze the prevalence levels for significant independence (http://www.socscistatistics.com). The significance threshold was set at $P = 0.05$. Ninety-five percent confidence intervals (CI) were calculated using an online calculator (http://epitools.ausvet.com.au).

Results

A total of 964 ear biopsies, 871 mites, 667 ticks and 743 fleas were analyzed individually for the presence of rickettsial species. Ear samples originated from the following rodents species: *A. agrarius*, *A. flavicollis*, *My. glareolus*, *M. arvalis*, *M. subterraneus* and *Mi. minutus*, and following insectivores: *Crocidura suaveolens* Pallas, 1811; *Neomys fodiens* Pennant, 1771; and *Sorex minutus* Linnaeus, 1776 and *S. araneus*. From the trapped small mammals, we identified the following mite species: *Laelaps agilis*, Koch, 1836; *Laelaps jettmari*, Vitzthum, 1930; *Laelaps hilaris*, Koch, 1836; *Androlaelaps fahrenholzi* (Berlese, 1911); *Hyperlaelaps microti*, Ewing, 1933; *Myonyssus gigas* Oudemans, 1912; *Haemogamasus nidi*, Michael, 1892; *Haemogamasus hirsutosimilis*, Willmann, 1952; *Haemogamasus hirsutus*, Berlese, 1889; *Eulaelaps stabularis*, Koch, 1836; *Euryparasitus marginatus*, Koch, 1939; *Eviphis ostrinus* (Koch, 1836); and two tick species—*I. ricinus* and *Ixodes trianguliceps*, Birula, 1895; and the following species of fleas: *Atyphloceras nuperus* (Jordan, 1931); *Amalaraeus penicilliger* (Grube, 1851); *Ctenophthalmus agyrtes*, Heller, 1986; *Ctenophthalmus assimilis* (Taschenberg, 1880); *Ctenophthalmus bisoctodentatus*, Kolenati, 1863; *Ctenophthalmus solutus*, Jordan et Rothschild, 1920; *Ctenophthalmus uncinatus* (Wagner, 1898); *Hystrichopsylla orientalis* (Smit, 1956); *Megabothris turbidus*, Rothschild, 1909; *Nosopsyllus fasciatus* (Bosc, 1801); *Peromyscopsylla bidentata* (Kolenati, 1860); and *Rhadinopsylla integella*, Jordan et Rothschild, 1921.

DNA of *Rickettsia* spp. was found in 4.6% (95% CI, 3.2–5.9) ear biopsies (Table 1) with the highest prevalence in *A. agrarius* (5.9%; 95% CI, 3.4–8.4), followed by *My. glareolus* (4.1%; 95% CI, 1.7–6.5) and the lowest prevalence was recorded in *A. flavicollis* (3.9%; 95% CI, 1.7–6.0) (see Table 1). Statistics were not performed when less than 20 individuals from the same species were collected. The prevalence of *Rickettsia* did not differ significantly among rodent species in Čermeľ ($\chi2 = 2.6$; $P = 0.2677$; $df = 2$) and Rozhanovce ($\chi2 = 3.4$; $P = 0.185$; $df = 3$). The total *Rickettsia* prevalence in small mammals did not differ significantly among localities ($\chi2 = 4.2$; $P = 0.237$; $df = 3$). Sequencing and/or *R. helvetica* specific real-time PCR showed the presence of *R. helvetica* in five samples from *A. agrarius*, three from *A. flavicollis*, one from *M. arvalis* and three *My. glareolus* (Acc No. KX051333). Furthermore, four sequences from *A. agrarius* (Acc. No. KX051329) and four from *A. flavicollis* showed 99.3–100% similarity with an uncultured *Rickettsia* sp. clone MG116–4 (Acc. No. DQ792806), (Choi et al. 2007). Two sequences obtained from *A. agrarius* (Acc. No. KX051330) were 99% similar with a *Rickettsia* endosymbiont of the *Dolichopus nubilus* strain 180 (Acc. No. JQ925593; Martin et al. 2013). Two sequences obtained from *My. glareolus* and two from *A. agrarius* (Acc. No. KX051331 and KX051332, respectively) were 87–90% similar with the *Rickettsia* endosymbiont *Chrysotus neglectus* (Wiedemann, 1817) (Acc. No. JQ925605, Martin et al. 2013). One ear sample from *My. glareolus* was amplified by *R. slovaca* specific real-time PCR and sequencing (21,958, Fig. S1) showed 100% identity with the *Rickettsia aeschlimannii* isolate Raeschlimannii_Tick15 identified in *Hyalomma marginatum* collected in China (Acc. No. MH267736), the uncultured *Rickettsia* sp. isolate Altai2–20,156-7 identified in *Dermacentor nuttalli* in the Altai (Acc. No. MF981014), and the *R. raoultii* isolate 109 identified in *D. reticulatus* in West Siberia (Acc. No. MH373353), but *R. raoultii* was not confirmed in any ear samples.

The total prevalence of rickettsiae in tested mites was 9.3% (95% CI, 7.4–11.2), (Table 2). The highest prevalence (51.9%; 95% CI, 33.0–70.7) was recorded in *L. jettmari*, followed by *H. nidi* (12.9%; 95% CI, 5.0–20.7), *L. agilis* (7.3%; 95% CI, 5.4–9.2) and *L. hilaris* (6.9%; 95% CI, 0–16.7) except species with low number of tested mites below 20. *Laelaps agilis* was the only *Rickettsia*-positive species in all study localities with significantly different occurrence among localities ($\chi2 = 25.0$; $P = 0.000$; $df = 3$). The prevalence of *Rickettsia* differed significantly between mites species in Rozhanovce ($\chi2 = 11.3$; $P = 0.004$; $df = 3$). Species specific real-time PCRs and sequencing showed the presence of *R. helvetica* in two *L. jettmari*, seven *H. nidi* and five *L. agilis*. Moreover, sequencing two *L. agilis* samples (Fig. S1) showed that sequences of Lagi461 and Lagi567 were 100% and 99% identical with *R. helvetica* (Acc. Nos. MG190375, MG190376, MF6473863) previously identified in Slovakia (Heglasová et al. 2018; Chvostáč et al., 2018), respectively. The presence of *R. slovaca* was confirmed in sixteen *L. agilis*, four *L. jettmari*, and one *E. stabularis*. Furthermore, sequences from *L. agilis* (Lagi388, Lagi392, Fig. S1) were 100% identical with *Rickettsia slovaca* isolate 2352AFLA (Acc. No. MG190377) previously identified in ear biopsy of *A. flavicollis* trapped in Eastern Slovakia

Table 1 Presence of *Rickettsia* spp. in skin biopsies of small mammals trapped in Eastern Slovakia between 2011 and 2013

Lokality Species	Botanical garden	Čermeľ	Hýľov	Rozhanovce	Total
Rodents					
AA	0/51/0	2/22/9.1 [0–21.1]	2/8/25 [0–55.0]	16/259/6.2 [3.3–9.1]	20/340/5.9 [3.4–8.4]
AF	1/22/4.6 [0–13.3]	1/65/1.5 [0–4.5]	6/55/10.9 [2.7–19.2]	4/170/2.4 [0.1–4.6]	12/312/3.9 [1.7–6.0]
MyG		5/115/4.4 [0.6–8.1]	0/39/0	6/113/5.3 [1.2–9.4]	11/267/4.1 [1.7–6.5]
MA		0/1/0		1/13/7.7 [0–22.2]	1/14/7.1 [0–20.6]
MS	0/1/0	0/2/0	0/2/0	0/2/0	0/7/0
MiM	0/1/0			0/2/0	0/3/0
Insectivores					
CS				0/2/0	0/2/0
NF		0/2/0			0/2/0
SA	0/1/0	0/6/0	0/4/0	0/4/0	0/15/0
SM		0/1/0		0/1/0	0/2/0
Total	1/76/1.3 [0–3.9]	8/214/3.7 [1.2–6.3]	8/108/7.4 [2.5–12.4]	27/566/4.8 [3.0–6.5]	44/964/4.6 [3.2–5.9]

No. of *Rickettsia*-positive/No. of tested samples/ Prevalence of *Rickettsia* in tissue samples of small mammals (%) [95% CI]

AA Apodemus agrarius; *AF Apodemus flavicollis*; *MyG Myodes glareolus*, *MA Microtus arvalis*; *MS Microtus subterraneus*; *MiM Micromys minutus*; *CS Crocidura suaveolens*; *NF Neomys fodiens*; *SA Sorex araneus*; and *SM Sorex minutus*

(Heglasová et al. 2018). Finally, sequences Lagi506 (Fig. S1) and *R. slovaca* isolate 2352AFLA (Acc. No. MG190377) were 98.2% similar. In these three sequenced samples, the presence of *R. slovaca* was confirmed by *R. slovaca*-specific real-time PCR. *Rickettsia* species were not identified in *L. agilis*, one *E. stabularis*, two *L. hilaris*, two *H. nidi* and three *L. jettmari*. Sequences from *L. agilis* (Lagi590) and *L. hilaris* (Lhil382) were 99.7% identical with the Israeli tick typhus rickettsia strain PoHu8122 (Acc. No. DQ821859) (Fig. S1). Sequences from *L. jettmari* (Ljett612) were 100% identical with the uncultured *Rickettsia* sp. isolate Altai2–2016-3 (MF981013) identified in *Dermacentor nuttalli* ticks from Russia. Sequences from *L. jettmari* (Ljeff378, Ljeff512) and Ljeff379 samples were 100% and 99.7% identical with sequences of the uncultured *Rickettsia* sp. clone MG116–4 (Acc. No. DQ795806) previously identified in mites (*Trombiculidae*) from Korea (Choi et al. 2007), respectively. *Rickettsia*-positive mites were collected from *Rickettsia*-

Table 2 Presence of *Rickettsia* spp. in mites (Mesostigmata) collected from small mammals trapped in Eastern Slovakia between 2011 and 2013

Locality Species	Botanical garden	Čermeľ	Hýľov	Rozhanovce	Total
LA	13/90/14.4 [7.2–21.7]	12/62/19.4 [9.5–29.2]	3/56/5.4 [0–11.3]	24/502/4.8 [2.9–6.7]	52/710/7.3 [5.4–9.2]
LJ	6/6/100	0/4/0		8/17/47.1 [23.3–70.8]	14/27/51.9 [33.0–70.7]
LH			0/1/0	2/28/7.1 [0–17.3]	2/29/6.9 [0–16.7]
AF				0/1/0	0/1/0
HM			0/5/0		0/5/0
MG	0/4/0	0/2/0	0/4/0	0/4/0	0/14/0
HN	1/16/6.3 [0–18.1]	0/4/0	0/2/0	8/48/16/7 [6.1–27.2]	9/70/12.9 [5.0–20.7]
HHs			0/2/0	0/1/0	0/3/0
HH				0/2/0	0/2/0
ES	4/6/66.7 [29–104.4]	0/1/0			4/7/57.1 [20.5–93.8]
EM	0/1/0			0/1/0	0/2/0
EO	0/1/0				0/1/0
Total	24/124/19.4 [12.4–26.3]	12/73/16.4 [7.9–24.9]	3/70/4.3 [0–9.0]	42/604/7.0 [4.9–9.0]	81/871/9.3 [7.4–11.2]

No. of *Rickettsia*-positive/No. of tested samples/ Prevalence of *Rickettsia* in mites from small mammals (%) [95% CI]

LA Laelaps agilis; *LJ Laelaps jettmari*; *LH Laelaps hilaris*; *AF Androlaelaps fahrenholzi*; *HM Hyperlaelaps microti*; *MG Myonyssus gigas*; *HN Haemogamasus nidi*; *HHs Haemogamasus hirsutosimilis*, *HH Haemogamasus hirsutus*; *ES Eulaelaps stabularis*; *EM Euryparasitus marginatus*; *EO Eviphis ostrinus*

Table 3 Presence of *Rickettsia* spp. in immature *I. ricinus* ticks from small mammals trapped in Eastern Slovakia between 2011 and 2013

Locality Host Species	Botanical garden	Čermeľ	Hýľov	Rozhanovce	Total
AA	29(24 + 5)/205/14.1 [9.4–18.9]	0/1/0.0		12(11 + 1)/96/12.5 [5.9–19.1]	41/302/13.6 [9.7–17.4]
AF	16(14 + 2)/108/14.8 [8.1–21.5]	2(2 + 0)/10/20.0 [0–44.8]	5(5 + 0)/30/16.7 [3.3–30.0]	17(17 + 0)/90/18.9 [10.8–27.0]	40/238/16.8 [12.1–21.6]
MyG		4(4 + 0)/8/50.0 [15.4–84.7]	10(10 + 0)/19/52.6 [30.2–75.1]	1(1 + 0)/23/4.3 [0–12.7]	15/50/30.0 [17.3–42.7]
MS	10(10 + 0)/35/28.6 [13.6–43.5]	1(1 + 0)/1/100.0	0/1/0.0	0/1/0.0	11/38/28.9 [14.5–43.4]
NF		1(1 + 0)/1/100.0			1/1/100.0
SA	1(1 + 0)/2/50.0	0/2/0.0	0/1/0.0		1/5/20.0
Total	56/350/16.0 [12.2–19.8]	8/23/34.8 [15.3–54.3]	15/51/29.4 [16.9–41.9]	30/210/14.3 [9.6–19.1]	109/634/17.2 [14.3–20.1]

No. of *Rickettsia*-positive (No. of positive larvae + No. of positive nymphs)/No. of tested *I. ricinus* ticks/ Prevalence of *Rickettsia* in *I. ricinus* from small mammals (%) [95% CI]

AA Apodemus agrarius; *AF Apodemus flavicollis*; *MyG Myodes glareolus*; *MS Microtus subterraneus*; *NF Neomys fodiens*; *SA Sorex araneus*

negative rodents, except in one instance with one *Rickettsia* sp.-positive *L. hilaris* fed on one *Rickettsia* sp.-positive *My. glareolus*.

In total, 634 *I. ricinus* and 33 *I. trianguliceps* ticks were analyzed for the presence of *Rickettsia* species. The overall prevalence of rickettsial infection in *I. ricinus* ticks was 17.2% (95% CI, 14.3–20.1), 93.5% positive ticks were larvae and 6.5% were nymphs (Table 3). The prevalence of *Rickettsia* positive *I. ricinus* ticks differed significantly between localities ($\chi 2 = 11.9$; $P = 0.007$; df = 3). The differences between *Rickettsia* positive ticks from the most abundant *A. agrarius* and *A. flavicollis* species were not statistically significant ($\chi 2 = 1.44$; $P = 0.229$; df = 1). The sequencing of 10 randomly selected ticks showed the presence of *R. helvetica* (Acc. No. KX051333). Two *R. helvetica*-positive ticks were collected from hosts, which were rickettsia negative, while all the others were from *Rickettsia*-positive hosts. In one sample, the sequencing showed 100% identity with *Candidatus Rickettsia mendelii* (Acc. No. KJ882305) identified in *I. ricinus* ticks collected from *Luscinia megarhynchos* in the Czech Republic (Hajduskova et al. 2016). In one sample, *I. ricinus*, sequences were identical with sequences from *My. glareolus* (21,958, Fig. S1) described above. In *I. trianguliceps* ticks, no rickettsial DNA was found.

DNA of *Rickettsia* spp. was found in 26 fleas, with a prevalence of 3.5% (95% CI, 2.2–4.8), (Table 4). In males *C. solutus* (Acc. No. KX179466) and *M. turbidus*, we found sequences 99.7% identical with *R. helvetica* (Acc. Nos. MG190375, MG190376, MF6473863). The respective hosts (*A. flavicollis* and *My. glareolus*) of these fleas were rickettsia negative. *Rickettsia* sp. originating from *C. assimilis* (Acc. No. KX179465) was 99.7% similar with the *Rickettsia* endosymbiont of *Hyalomma marginatum* (Acc. No. KC689908) identified in *H. marginatus* infested humans from Turkey. Its host, *M. arvalis* was *Rickettsia* negative. Three sequences of *Rickettsia* (Acc No. KX179467) from two females and one male of *M. turbidus* from rickettsia-negative *A. agrarius*, *A. flavicollis* and *My. glareolus* were 95% similar to *Rickettsia* endosymbiont of *Nesidiocoris tenuis* (Acc. No. KF646706) originated from the mirid bug *N. tenuis* identified in Israel (Caspi-Fluger et al. 2014). *Rickettsia* spp. originated from *C. assimilis* fleas were also identical with sequences from *My. glareolus* (21,958, Fig. S1) described above.

Discussion

Small mammals are important hosts for immature stages of ticks and are considered reservoir hosts for some tick-borne pathogens, such as tick-borne encephalitis virus, *Borrelia afzelii*, *Borrelia garinii*, *Borrelia miyamotoi*, *Babesia microti* and *Anaplasma phagocytophilum* (Randolph et al. 1999; Humair et al. 1999; Huegli et al. 2002; Burri et al. 2014;

Obiegala et al. 2015). Evaluating the role of rodents in rickettsia life cycles has been the subject of many studies (Sprong et al. 2009; Svoboda et al. 2014; Miťková et al. 2015; Špitalská et al. 2015; Obiegala et al. 2016; Minichová et al. 2017). In this study, we showed the presence and the molecular characterization of the following rickettsial species: *R. helvetica*, *R. slovaca*, unidentified *Rickettsia* and rickettsial endosymbionts in eight species of ectoparasites, tick, fleas or mites, (*L. jettmari*, *H. nidi*, *L. agilis*, *E. stabularis*, *I. ricinus*, *C. solutus*, *C. assimilis and M. turbidus*) infesting four species of small mammals (*A. agrarius*, *A. flavicollis*, *M. arvalis* and *My. glareolus*) circulating in the Košice region, Eastern Slovakia. Considering the trophic relationships related to small mammals, two mite species, *L. jettmari* and *L. agilis*, are considered obligatory parasites living all their cycle on the body of rodents. The next two species, *H. nidi* and *E. stabularis*, represent facultative hematophagous parasites, in their life cycle with alternately living conditions in nests and on the host bodies (Mašán and Fend'a 2010). For fleas, only *C. assimilis* has stronger trophic relationships to *M. arvalis* or other microtid rodents. The next two rickettsia positive flea species, *C. solutus* and *M. turbidus*, infest a broader host spectrum of rodents including the *Apodemus*, *Myodes*, *Microtus* and *Micromys* genera (Rosický 1957). Similarly broad host parasitism is common in *I. ricinus* ticks (Fillipova, 1977). We reported the highest rickettsial infection in *I. ricinus* ticks followed by mites, and the lowest was recorded in fleas. A similar study was conducted in Lithuania by Radzijevskaja et al. (2018); ectoparasites collected from rodents were investigated for the presence of *Rickettsia* spp. They found the highest infection in fleas, followed by *I. ricinus* ticks and then mites. Results suggest that different vectors share small mammals as reservoir hosts and rickettsia-positive ectoparasites could be infected by the acquisition of bacteria via blood meals from rickettsiemic hosts, or by co-feeding of ectoparasites through infected blood meals. They could be also infected during previous feedings on another host. Furthermore, positive larvae could be the result of transovarial transmission from infected females or through transstadial transmission passed onto the next developmental stages.

Rickettsial species in ticks in Slovakia have been studied during the last decade, and the presence of *R. helvetica* and *R. slovaca* in *I. ricinus* and *Dermacentor* spp. was previously confirmed (Špitalská et al. 2018) and thus expected in ticks from this study. *R. helvetica* was confirmed not only in *I. ricinus* ticks but also in *L. jettmari*, *H. nidi* and *L. agilis* mite species and *C. solutus* and *M. turbidus* flea species. Previously, *R. helvetica* was identified in *C. agyrtes* males collected from *A. agrarius*, which was for the first time, it was identified in fleas from Slovakia in 2015 (Špitalská et al. 2015). Similarly, Sprong et al. (2009) identified *R. helvetica* in fleas isolated from *My. glareolus* and *Apodemus sylvaticus* from the Netherlands. *R. helvetica* was also identified in *Archaeopsylla erinacei* collected from hedgehogs and in *Nycteridopsylla eusarca* bat fleas from Hungary (Hornok et al. 2012, 2014). The presence of *R. helvetica* in pooled samples of *L. agilis*, *H. nidi* (Laelapidae), *Hirsutiella zachvatkini* and *Kepkatrombicula štorkáni* (Trombiculidae) mites collected from small terrestrial mammals captured in south-western Slovakia was confirmed

Table 4 Presence of *Rickettsia* spp. in fleas (Siphonaptera) from small mammals trapped in Eastern Slovakia between 2011 and 2013

Locality Flea Species	Botanical garden	Čermeľ	Hýľov	Rozhanovce	Total
AN			0/2/0.0		0/2/0.0
AmP	0/1/0.0	0/2/0.0	0/1/0.0		0/4/0.0
CA		0/7/0.0	0/19/0.0	2/227/0.9 [0–2.1]	2/253/0.8 [0–1.9]
CAs				4/97/4.1 [0–8.1]	4/97/4.1 [0–8.1]
CB				0/1/0.0	0/1/0.0
CS		0/1/0.0	1/24/4.2	4/182/2.2 [0–4.3]	5/207/2.4 [0.3–4.5]
CU		0/5/0.0	0/7/0.0		0/12/0.0
HO				0/1/0.0	0/1/0.0
MT	0/3/0.0		1/1/100.0	14/117/12.0 [6.1–17.9]	15/121/12.4 [6.5–18.3]
NF		0/1/0.0		0/5/0.0	0/6/0.0
PB		0/8/0.0	0/19/0.0		0/27/0.0
RI		0/9/0.0	0/3/0.0		0/12/0.0
Total	0/4/0.0	0/33/0.0	2/76/2.6 [0–6.2]	24/630/3.8 [2.3–5.3]	26/743/3.5 [2.2–4.8]

No. of *Rickettsia*-positive/No. of tested fleas/Prevalence of *Rickettsia* in fleas from small mammals (%) [95% CI]

AN Atyphloceras nuperus; *AmP Amalaraeus penicilliger*; *CA Ctenophthalmus agyrtes*; *CAS Ctenophthalmus assimilis*; *CB Ctenophthalmus bisoctodentatus*; *CS Ctenophthalmus solutus*; *CU Ctenophthalmus uncinatus*; *HO Hystrichopsylla orientalis*; *MT Megabothris turbidus*; *NF Nosopsyllus fasciatus*; *PB Peromyscopsylla bidentate*; *RI Rhadinopsylla integella*

by Miťková et al. (2015). Radzijevskaja et al. (2018) mentioned the presence of *R. helvetica* in *I. ricinus* ticks, in *L. agilis* and *M. gigas* mites and in *C. agyrtes*, *M. turbidus* and *M. walkeri* fleas.

The natural known vectors of *R. slovaca* are *Dermacentor* species ticks, and in this study, such pathogen was confirmed in *L. agilis*, *L. jettmari* and *E. stabularis*, which shows that it can be acquired from other hosts. Previously, *R. slovaca* was identified in *I. ricinus* in Austria (Schötta et al. 2017) and in *Melophagus ovinus*, the sheep ked in China (Liu et al. 2016). The presence of unidentified *Rickettsia* species or rickettsial endosymbiont in different species of ectoparasites has been recorded in many studies (Blair et al. 2004; Khrouf et al. 2014; Segura et al. 2014; Song et al. 2018), and it could be result of mis-identity with multiple *Rickettsia* species, unsuccessful amplification of other genes, low DNA quantity used for other amplifications, low quality of DNA, as usually the isolate is not obtained by culture, thus the characterization is not complete and the amount of DNA based on what is collected and extracted.

The low prevalence of *Rickettsia* in skin biopsies was expected, as suggested by the previous study of Minichová et al. (2017). The presence of different rickettsial species was expected. However, we expected also the presence of *R. monacensis* because this species has been identified in previous studies from Slovakia, Lithuania and Italy (Špitalská et al. 2014, Špitalská et al. 2016, Minichová et al. 2017, Radzijevska et al. 2018, Mori et al. 2018). We assumed that no identification of *R. monacensis* could be caused by not all positive samples were sequenced and partially due to the competition with the most prevalent *R. helvetica* in Slovakia (Švehlová et al. 2014; Špitalská et al. 2014; Špitalská et al. 2016).

Finally, the present data extend the knowledge on the distribution of *Rickettsia* spp. and their association with different arthropod vectors and indicate the presence of phylogenetically distinct rickettsiae in multiple ectoparasite species feeding on different rodent hosts. Results confirmed the high circulation of bacteria from the genus *Rickettsia* in mites, ticks and fleas collected from small mammals in Košice region, Eastern Slovakia, with related clinical cases of rickettsiosis. Some mite, flea and tick species infesting wild rodents might carry *Rickettsia* species, and small mammals can serve as carriers of *Rickettsia* infected ectoparasites.

Conclusions

We are reporting here the occurrence and molecular characterization of *Rickettsia* spp. in eight species of ectoparasites found on four species of small rodents in Eastern Slovakia. The data presented in this paper extend the knowledge on the distribution of *Rickettsia* species, their association with different arthropod vectors and rodent-parasites interactions.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All applicable international, national, and/or institutional guidelines for the care and use of animals were followed.

All procedures performed in studies involving animals were in accordance with the ethical standards of the institution or practice at which the studies were conducted.

Small mammals were euthanized according to the laws of the Slovak Republic under the licences of the Ministry of Environment of the Slovak Republic No. 297/108/06–3.1 and No. 6743/2008–2.1.

This article does not contain any studies with human participants performed by any of the authors.

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